

LAB 3

Serial Interface-UART

Objectives:

WEEK 10

- To write a program to transfer data to a PC serially.
- To write a program to receive data from a PC serially.

Material:

- Keil IDE
- TI Tiva development board
- A Terminal software like Tera Term or Putty.
- Wires
- A computer

Submission Instructions:

- Demonstrate your work for each activity to your instructor. Missing the instructor signature results in 20% penalty.
- Submit a zipped file of your project folder onto the BB dropbox. Missing the zipped project file results in a 20% penalty.
- Write a lab report detailing the implementation and results. Include the schematic and picture of your circuit, screenshot of your code and results. Answer all the questions.

NOTE:

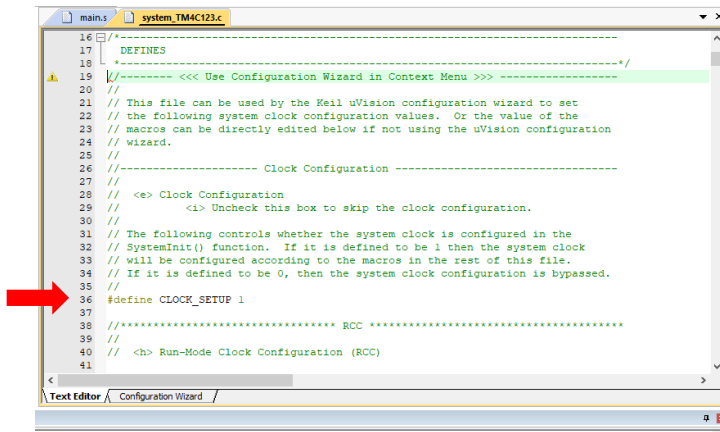
- All answers submitted for this lab must be original and based on your own experimental work and observations.
- Your explanations must reflect what you observed during your lab session.
- Do not copy answers from classmates, previous reports, or online sources.
- Generic or theoretical answers that do not match your actual experimental results will not receive full credit.
- Be prepared to explain your answers verbally if requested.
- Your grade will be based not only on correctness, but also on your ability to demonstrate genuine understanding of your implementation and results.

Pre-Lab Activity 1

Install a terminal software such as Tera Term or Putty on your computer.

Pre-Lab Activity 2

In **system_TM4C123.c** file, change **#define CLOCK_SETUP 1** to **#define CLOCK_SETUP 0**.



```
16 //-----
17 #define
18 //-----
19 //----- <<< Use Configuration Wizard in Context Menu >>> -----
20 //
21 // This file can be used by the Keil uVision configuration wizard to set
22 // the following system clock configuration values. Or the value of the
23 // macros can be directly edited below if not using the uVision configuration
24 // wizard.
25 //
26 //----- Clock Configuration -----
27 //
28 // <e> Clock Configuration
29 // <i> Uncheck this box to skip the clock configuration.
30 //
31 // The following controls whether the system clock is configured in the
32 // SystemInit() function. If it is defined to be 1 then the system clock
33 // will be configured according to the macros in the rest of this file.
34 // If it is defined to be 0, then the system clock configuration is bypassed.
35 //
36 #define CLOCK_SETUP 1
37 //----- RCC -----
38 //
39 // <h> Run-Mode Clock Configuration (RCC)
40 //
41
```

Lab Activity 1 [10 points]

1. Connect your trainer board to the PC's USB port. (Use a USB to serial convertor to connect the TXD and RXD pins of the microcontroller to the USB port of your PC.
2. Configure Terminal program and serial port for 9600 baud rate, 8-bit , no parity, 1-stop bit, no FIFO.
3. Copy and paste “Program 4-1”, compile and upload it onto your microcontroller. This program will send the message “YES” continuously to PC.
4. Modify your code for a baud rate of 115200 and confirm that the message “YES” is continuously sent to PC.
5. Demonstrate your work to your instructor.

Lab Activity 2 [25 points]

1. Configure your terminal program with the following settings:

Baud rate: 9600

Data bits: 8

Parity: None

Stop bits: 1

Flow control: None

2. Load and Test Program 4-2

- Copy and paste **Program 4-2** into your IDE.
- Compile and download it to the TM4C123 microcontroller.
- Run the program.

This program:

- Receives characters from **UART0**
- Displays the **binary value (ASCII code)** of the received character on the tri-color LEDs connected to **Port F (PF3–PF1)**.

Test Procedure:

- In the terminal, press any key:
 - Uppercase letters (A–Z)
 - Lowercase letters (a–z)
 - Numbers (0–9)

Observe that the ASCII value of the pressed key is displayed in binary on the LEDs.

- Type the following characters in order:
A, t, 5, ?
Record the exact LED pattern observed for each.
Explain why each LED pattern corresponds to the ASCII value.

3. Modify the program so that:
 - The received character is transmitted back to the serial terminal.
 - The terminal displays the same character you typed (echo behavior).
Hint: You may need to refer to Program 4.1
4. Modify your code for the following control sequence:
 - If user types r → turn on Red LED (PF1)
 - g → green LED
 - b → blue LED
 - o → all off

- h → print menu again
- * → print a funny secret message
- Any other input → print “Invalid command”

Demonstrate your work to your instructor.

Lab Activity 3 [20 points]

Modify your code in Lab Activity 1 to:

1. Add a function `Void UART0Tx_String(char str[])` . This function receives any string as an argument and transmits it to the terminal via UART0.
2. In `main()`, call your function with any message of your choice, for example:
`UART0_Tx_String("Hello from TM4C123!\r\n");`

Lab Activity 4 [20 points]

Modify your code to:

1. Add a function `Void UART0Rx_String(char str[])` . This function receives a complete string from the serial terminal via UART0.
2. In the main function:
 - Declare a character array (buffer) of appropriate size (e.g., 20 characters).
 - Prompt the user to enter a string.
 - Call your `UART0_Rx_String()` function.
 - Display the received string back to the terminal.

Testing Procedure

1. Open the serial terminal (9600 baud, 8N1).
2. Type any word or short sentence.
3. Press Enter.
4. Verify that:
 - Your typed text is echoed.
 - The program displays the full string correctly.

Demonstrate your work to your instructor.

Questions [25 points]

1. [3 point] Why do we configure UART with parameters such as 8-bit data, no parity, and 1 stop bit?
2. [3 point] What happens if the baud rates of the sender and receiver do not match?
3. [3point] How did you verify that the correct data was being transmitted and received? If you encountered errors in communication, what debugging steps did you take?
4. [3 point] When you configured 9600 baud, what exact COM port number appeared on your computer? Provide a screenshot and explain how you identified it.
5. [3 point] Highlight the exact line in your code that:
 - Enables UART0
 - Enables TX
 - Enables RX

Explain what each bit does.

6. [3 point] Disconnect and reconnect the board while terminal remains open.
 - What happens to the COM port?
 - Does the program restart automatically?
 - Why?
7. [7 point] Measure the time required to transmit 100 characters at 9600 baud and 8 data bits. Compare the measured time with the theoretical calculation. Show all steps of your work for calculations and your modified code. Explain any difference.